Paying Attention:
Overnight Returns and the Hidden Cost of Buying at the Open

March 2011

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*Corresponding author. We acknowledge the helpful comments of Chris Anderson, Audra Boone, Bob DeYoung, Laura Field, Ben Jacobsen, Kelly Welch, Jide Wintoki, and seminar participants at the University of Auckland, Massey University, the University of Kansas, the New Zealand Finance Colloquium, the SEC, and the national conferences of the Financial Management Association and the Eastern Finance Association. We thank Jeff Harris, Frank Hatheway, and Nasdaq-OMX for access to proprietary data. We also thank Xin Zhao for excellent research assistance. In addition, Koch acknowledges support from the University of Auckland and Massey University where he served as visitor while conducting this research. Tuttle acknowledges research support from the Commodity Futures Trading Commission where she served as a visitor.

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Abstract

Using 13 years of intraday data for U.S. stocks, we find a strong tendency for positive returns during the overnight period followed by reversals during the trading day. This behavior is driven by an opening price that is high relative to intraday prices. We find this temporary price inflation at the open is concentrated among stocks that have recently attracted the attention of retail investors, and these high attention stocks have high levels of net retail buying at the start of the trading day. In addition, we document that the sensitivity of opening prices to retail investor attention is more pronounced for stocks that are difficult to value and costly to arbitrage, and is greater during periods of high overall retail investor sentiment. The additional implicit transaction costs for retail traders who buy high attention stocks near the open frequently exceed the effective half spread.

JEL Classification: D82, G14, G19.
Key Words: market efficiency, attention, sentiment, retail investors, transaction costs, short sale restrictions, institutional ownership, opening price, overnight return.

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1. Introduction

Behavioral finance theories assume that individual investors are subject to sentiment that makes them willing to trade at prices not justified by fundamentals. Moreover, sentiment-driven retail traders tend to act as a group, trading the same stocks at the same time and in the same direction. Because trading against these investors can be costly and risky, the collective price impact of their trading can be substantial, and sentiment-based trading can cause prices to deviate from fundamental value for long periods of time.[1]

Barber and Odean (2008) contribute to this literature by theorizing that retail investors herd into attention-grabbing stocks. They observe that retail investors who want to buy face a different search problem than those who want to sell. When retail investors want to buy, they must select among thousands of stocks. Odean (1999) and Barber and Odean (2008) hypothesize that individual investors manage this problem by limiting their search to stocks that have recently attracted their attention. However, when retail investors want to sell, their decision of which stock to sell is normally limited to the small set of stocks they own, since retail investors typically do not short sell. Consistent with this theory, Barber and Odean (2008) show that individual investors are net buyers on the next trading day following days with high absolute returns, which is one of their proxies for attention.

We extend the theory of Barber and Odean (2008) to develop several predictions with regard to intraday patterns in retail order flow and price formation. Our first prediction is that an attention-triggering event on day *t* will lead to retail buying near the open on day *t+1*, because the open is the first opportunity since the previous close for retail investors to buy these high

1 For theoretical work in this area, see DeLong et al. (1990) and Shleifer and Vishny (1997). Kumar and Lee (2006) provide empirical evidence that herding by retail traders helps to explain returns for stocks with high retail concentration that are also difficult to arbitrage. Barber, Odean, and Zhu (2009) show that stocks bought by retail investors underperform stocks sold by retail investors over the following year. Baker and Wurgler (2006) find that, after periods of high (low) sentiment, stocks that are difficult to value and costly to arbitrage earn low (high) returns.

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attention stocks.[2] Second, we hypothesize that this attention-driven retail buying pressure results in opening prices that are high relative to prices during the rest of the trading day, especially for stocks that are difficult to value and costly to arbitrage. Finally, we expect that the impact of attention on opening prices is greater during periods of high overall retail investor sentiment.

We test these predictions for the 3,000 largest U.S. stocks over the period, 1996-2008. In preliminary analysis, we examine quote midpoints at the open and close, and find significant positive mean overnight returns of +10 basis points (bp) per day, along with negative trading day reversals of -7 bp per day. We then explore whether these overall tendencies are due to a high opening price, or a low closing price, or both. Consistent with our theory of attention-based overpricing at the open, we find the opening price is high relative to subsequent intraday prices, while there is no tendency for the closing price to decline further below intraday prices.[3]

These descriptive results are consistent with evidence in two papers that were developed simultaneously with ours. Branch and Ma (2008) find a negative correlation between the overnight return and the subsequent trading day return for U.S. stocks. They suggest that this tendency may relate to the microstructure of how specialists and market-makers behave at the open. Cliff et al. (2008) find evidence that the U.S. equity premium over the last decade is solely due to positive overnight returns. They observe that evidence of systematic negative daytime reversals "present(s) a serious challenge to traditional asset pricing models from the standpoint that these models do not predict negative average returns." Both of these studies join our work to emphasize that this behavior represents a surprising new anomaly, and call for further efforts to find an explanation. We extend the analysis, and show that our theory of attention-triggered retail buying at the start of the trading day offers a unifying explanation for this behavior.

2 There is an after-hours market and a pre-open on ECNs, but high costs and other impediments to trading deter retail investors, so that professional traders dominate these markets (see Barclay and Hendershott 2003, 2004).
3 Reliance on midquotes ensures that our results are not due to bid-ask bounce. Trade prices yield similar results.

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Empirical analysis of our theory requires a proxy for the level of retail investor attention at the start of the typical trading day. We use two size-adjusted market-based variables from the previous day. First, following Barber and Odean (2008), we consider the squared return yesterday as a proxy for news that could attract the attention of retail investors today.[4] Second, motivated by the main result in Barber and Odean that individual investors are net buyers of attention-grabbing stocks, we use actual net buying by individual investors all day yesterday as a percent of total volume, to proxy for retail attention at the start of trading today.

We obtain this second proxy using proprietary data on the intraday trading activity of retail investors, for an abridged sample of Nasdaq stocks during 1997-2001.[5] These data also enable measurement of the intensity of net buying by retail investors near the open, relative to that during the rest of the trading day. Consistent with our first hypothesis, we find that: (i) the intensity of retail buying near the open is significantly greater for high attention stocks than for low attention stocks, and (ii) the intensity of retail buying of high attention stocks is significantly greater during the first hour than it is during the rest of the trading day.

We next examine the implications of these intraday patterns in net retail buying for price formation at the open. When we analyze our main sample of the 3,000 largest U.S. stocks during 1996-2008, we find the subsample of high attention stocks has a significant mean overnight return (trading day reversal) of +13 (-13) basis points (bp) per day. In contrast, the analogous results for the subsample of low attention stocks are much smaller, at only +3 (-3) bp per day. When we examine the abridged sample of Nasdaq stocks during 1997-2001, the mean overnight return (trading day reversal) for high attention stocks is larger in magnitude, at +26 (-27) bp per

4 In addition to absolute returns yesterday, Barber and Odean (2008) consider contemporaneous daily volume as an alternative measure of retail attention. We have also used share turnover yesterday to proxy for retail investor attention at today's open, and we find robust results.

5 We are grateful to Jeff Harris, Frank Hatheway, and Nasdaq OMX for providing these proprietary data. See Griffin et al. (2003) for details about the data.

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day. Once again, the analogous results for low attention Nasdaq stocks during this period are much smaller, at +4 (-12) bp per day. Importantly, the average 24-hour (close-to-close) return is never significantly different from zero for any subsample examined. Together, these results support our second hypothesis, indicating a significant price impact of attention-triggered retail buying pressure at the open, which is only a short-term intraday phenomenon.

Based on behavioral finance theories, we expect that stock prices may deviate further from fundamentals if the stock is more difficult to value and more costly to arbitrage.[6] We test this hypothesis by extending our analysis in two ways. First, we consider the role of institutional ownership in concert with our proxies for retail attention. Stocks with low institutional holdings have a high concentration of retail investors, and are difficult to short sell.[7] Thus, for these stocks, we expect greater upward price pressure at the open on days following news that attracts the attention of retail investors. Our results support this prediction. Now the subset of our main sample of stocks subject to both high attention and low institutional ownership has a larger mean overnight return (trading day reversal), at +17 (-17) bp per day. Similarly, the analogous stratification from our abridged sample of Nasdaq stocks also has a larger mean overnight return (trading day reversal), at +34 (-42) bp per day. In contrast, the subsample that is least prone to overpricing at the open, with low attention and high institutional ownership, has overnight returns and trading day reversals that lack economic or statistical significance.

Second, we show that these overnight return patterns increase in magnitude for finer subsamples of stocks that are more difficult to value and more costly to arbitrage. We analyze a subset of the main sample that includes only firms with high recent volatility, high transaction

6 For example, see Baker and Wurgler (2006) and Kumar and Lee (2006).
7 Kumar and Lee (2006) show that retail trading is negatively correlated with institutional ownership. In addition, a large body of work uses low institutional ownership to proxy for binding short sale constraints (e.g., see Almazan et al., 2004, Asquith et al., 2005, Berkman et al., 2009, D'Avolio, 2002, Nagel, 2005, and Ofek et al., 2004).

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costs, and high short interest. For this subset, the mean overnight return (trading day reversal) for the finer subsample of stocks subject to both high attention and low institutional ownership is +43 (-45) bp per day. Likewise, the analogous results for the abridged sample of Nasdaq stocks are +61 (-70) bp per day.

The high opening prices we find for stocks subject to high attention and low institutional ownership represent a substantial hidden cost of buying at the open. Our tests are predictive, and thus indicate that postponing purchases of these stocks from the open until later in the day can avoid these hidden costs. Similarly, selling these stocks at the open, rather than later in the day, can lead to major improvements in performance. For example, a mean trading day reversal of 45 bp per day accumulates to 112.5% per annum.

Finally, we investigate whether the magnitude of positive overnight returns and negative trading day reversals is exacerbated during periods with high overall investor sentiment (see Baker and Wurgler, 2006, 2007). We consider two time series proxies that measure variation over time in the level of retail investor sentiment. Our first measure is the sentiment index of Baker and Wurgler, while our second measure is based on market-wide net buying activity by retail investors during the first hour of the trading day. We find that variation over time in the magnitude of opening price inflation, for stocks subject to high attention and low institutional holdings, is significantly related to both of these time series indices of market sentiment. For example, mean overnight returns and trading day reversals are more than twice as large during months with high versus low sentiment. In addition, during periods of high investor sentiment, this opening price inflation can be twice the magnitude of the effective half spread. This evidence establishes that the behavior of overnight returns and trading day reversals documented in this study is both economically and statistically significant.

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Our results are robust when we apply this portfolio approach using market-adjusted returns, median returns, or trade prices to measure returns, and when we exclude low-price stocks. This behavior is also ubiquitous across small and large stocks, Nasdaq and NYSE stocks, growth and value stocks, and momentum winners and losers, although there is a larger mean overnight return and trading day reversal for small stocks, Nasdaq stocks, growth stocks, and momentum losers. We also find these patterns are largest on Mondays, but they also appear on the other days of the week. In addition, these results appear in all subperiods. Finally, we find similar results and conclusions when we apply an alternative Fama-MacBeth regression approach.

This article proceeds as follows. Section 2 describes our sample selection and variable construction. Section 3 reports descriptive statistics. Section 4 presents the main empirical analysis. Section 5 analyzes the time series relation between overall market sentiment and the magnitude of opening price inflation. Section 6 provides robustness tests, and section 7 summarizes and concludes.

2. Sample Selection and Variable Construction

2.1 Sample Selection and Daily Return Measures

Our main sample includes the 3,000 largest U.S. firms, according to their market capitalization on July 1 of each year over the period, 1996-2008. We also apply our analysis to an abridged sample for which proprietary data on intraday retail trading were obtained from Nasdaq OMX. These proprietary data describe each trade for Nasdaq stocks during the period, 1997-2001, and include the identity of market participants on each side of the trade. For this abridged sample, we follow the procedures in Griffin et al. (2003), and classify both sides of all trades as originating from an individual or an institution based on the market participants involved in the trade.

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We calculate daily returns using quotations from TAQ.[8] The opening price on day t ($open_t$) is the midpoint of the first valid bid and ask quotes after 9:30 a.m.[9] The closing price on day t ($close_t$) is the midpoint of the last valid bid and ask quotes before 4:00 p.m.[10] We adjust these daily opening and closing prices for stock splits and dividends, before computing daily returns.[11] Returns are measured as the log of the price relative over each time frame considered:

$$\begin{aligned}
&\text{24-Hour, Open-to-Open Return} && = oto_t = \log(open_t / open_{t-1});\\
&\text{24-Hour, Close-to-Close Return} && = ctc_t = \log(close_t / close_{t-1});\\
&\text{Overnight, Close-to-Open Return} && = cto_t = \log(open_t / close_{t-1});\\
&\text{Trading Day, Open-to-Close Return} && = otc_t = \log(close_t / open_t).
\end{aligned}$$

Note that $ctc_t = cto_t + otc_t$. Finally, consistent with prior work, we screen the data for errors and extreme observations.[12]

2.2 Variable Construction

For each stock, we estimate two proxies for the level of attention by retail investors at the start of the trading day. Our first proxy is the square of yesterday's close-to-close return (VOL_{t-1}). This measure is motivated by Barber and Odean (2008), who use the absolute return yesterday to

8 TAQ's consolidated quotation file is an aggregation of quotes within each market venue. It represents a set of "top of book" records for each venue. In order to identify market-wide best prices, we calculate an inside market across all venues. For the NYSE, this calculation almost never improves upon the NYSE specialist's price. However, in Nasdaq issues, ECNs often improve on the Nasdaq's reported best price.

9 By "the first valid quotes" we mean the first quotes after 9:30 a.m. for which there is non-zero trade size on both the bid and the ask. For NYSE issues, selection of the open is straightforward since the opening cross is easily identified and begins the trading day. For Nasdaq issues, selection of the open is not so straightforward. Although technically the Nasdaq opens at 9:30 a.m., it is often several minutes before valid market-maker quotes appear. If the midquote precisely at 9:30 a.m. is used as the open, these quotes will often be flagged as "closed" by the market participant, or they may have zero size associated with the prices. In either case the price does not represent a firm commitment to trade. This behavior of the Nasdaq open motivates our choice of the first valid quotes after 9:30 a.m. as the opening quotes. Our results are robust when we take the midquote precisely at 9:30 a.m. as the open.

10 Occasionally, the final quotes before 4:00 p.m. have zero shares available to trade on one or both sides, or the quote is flagged as closed. For this reason, we select the last valid quote before 4:00 p.m.

11 On days when a cash dividend or a stock split becomes effective at the open, we adjust the TAQ data on opening and closing prices using CRSP data on the amount of the cash dividend or the multiple of the stock split. The adjusted opening and closing prices are then used to compute each daily return measure.

12 Quotes are dropped from our analysis if their "mode" designation indicates that they are not normal quotes, or if the reported ask price is greater than 1.5 times the reported bid price. In addition, we omit the daily return if: (i) the bid-ask spread at the open or close is negative, or (ii) the opening or closing spread is greater than $5.00 or 30% of the midpoint quote, or (iii) the effective half spread is greater than $2.50 or 15% of the midpoint quote, or (iv) the daily open is more than 25% greater (or less) than both the previous close and the subsequent close. We also drop all daily returns for a firm if daily quotes are missing for more than 25 percent of all trading days for that firm.

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proxy for news that could attract the attention of retail investors, and find that increases in this measure are associated with increases in net retail buying throughout today. Our second proxy is net shares bought by individual investors yesterday, as a percent of total daily share volume ($Retail_NetBuy_{t-1}$). This measure is also motivated by the results in Barber and Odean. Their finding that individual investors are net buyers of attention-grabbing stocks suggests that stocks with high net retail buying have (almost by definition) attracted the attention of retail investors.

We also use the abridged sample of Nasdaq stocks over 1997-2001, to compute three different measures of the intensity of net buying by retail investors near the day's open:

$NetBuy_Open_t$ = the mean proportion of stocks on day t for which the first trade of the day is a purchase by a retail investor, minus 0.5;

$NetBuy_15Min_{it}$ = (Net Retail Buy Volume in 1st 15 min) / (Total Retail Volume in 1st 15 min);

$NetBuy_DIFF_{it}$ = [(Net Retail Volume in 1st hour) – (Net Retail Volume in last 5.5 hours / 5.5)] / (shares outstanding);

where Net Retail Buy $Volume_{it}$ = the net number of shares of stock i bought by retail investors over each time frame, on day t;

and Total Retail $Volume_{it}$ = the total number of shares of stock i traded by retail investors over each time frame, on day t.

The first variable defined above measures the 'net' tendency for the first trade of the day to be a purchase by a retail investor. The next variable reflects the flow of net purchase volume by retail investors, as a percent of total retail share volume, over the first fifteen minutes of the trading day. The last variable captures the difference between the rate of net retail buying during the first hour and the hourly rate of net retail buying during the rest of the trading day, as a percent of shares outstanding.[13]

Next, following Asquith et al. (2005), we consider two proxies for short sale constraints: institutional ownership and relative short interest. Data on institutional holdings are from CDA

13 We have also analyzed net retail buy volume over the first minute and the first five minutes of the trading day. Analysis of these variables yields similar results to the measures defined here.

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Spectrum 13F filings. Each quarter, we compute the percentage institutional ownership ($INST_t$) for every firm as aggregate shares held by institutions scaled by shares outstanding. If a stock is available in CRSP but has no information on institutional holdings, we assume that the stock has zero institutional ownership (see Asquith et al., 2005, Gompers and Metrick, 2001, and Nagel, 2005). Our second proxy for short sale constraints is relative short interest (RSI_t), measured by the total number of shares sold short as a percent of shares outstanding, using monthly data on short interest from the NYSE and the Nasdaq.

Finally, we use three measures of transaction costs: the percentage spread at the open ($SPR(open)_t$), at the close ($SPR(close)_t$), and the percentage effective half spread ($Spread_t$).[14]

3. Descriptive Statistics and Intraday Price Patterns

3.1 Descriptive Statistics for Overnight Returns and Trading Day Returns

In Panel A of Table 1, we report the mean and median values for overnight returns, trading day returns, and 24-hour returns across all firms and days in the sample. Note that the average number of firms each day varies from 2,447 to 2,604 across the different measures of daily returns considered. This number is less than the 3,000 largest U.S. stocks used as the initial sample, because daily opening or closing quotes are sometimes invalid or unavailable for some of the smaller stocks (see footnote 12 for details of our screening procedure).

Standard t-tests applied to these mean returns could be biased due to cross-correlation of daily returns across firms on the same date (Bernard, 1987). In our analysis we adjust our standard errors for this possibility. Specifically, for each of the 3,268 trading days in our 13-year sample period, we first compute the cross-sectional mean (or median) overnight and trading day

14 For every trade, the percentage effective half spread is defined as the absolute difference between the trade price and the quote midpoint, as a percent of the quote midpoint. Trades are matched to quotes with a lag of one second, and then averaged to get the day's percentage effective half spread. Our choice of a one-second lag is taken from the Nasdaq Economic Research Office, which argues that this lag is optimal to match trades and quotes in its automated electronic system. Our results are not affected by this choice (see Lee and Ready, 1991, and Bessembinder, 2003).

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returns across all stocks in the sample. We then compute the time series average of these cross-sectional mean (or median) returns across all trading days in the sample period. The corresponding t-statistics are based on the standard errors of the time series mean returns.

Results on the left side of Panel A in Table 1 indicate a significant positive mean overnight return (cto) of approximately +10 basis points (bp) per day, and a significant negative trading day reversal (otc) of -7 bp per day, when averaged across all firms and days. As a result, the mean return on a strategy that is long the sample stocks during the overnight period and short the same stocks during the subsequent trading day (DIFF) would be +17 bp per day, before deducting transaction costs. On the other hand, when we subtract the average bid-ask spread at the daily open and close, we obtain a significant negative mean difference after transaction costs (DIFF-TC) of -71 bp. Note that the median value of the overnight return in Panel A of Table 1 (cto) is somewhat smaller than the mean, while the median trading day return in Panel A (otc) is approximately the same as the mean. Finally, the 24-hour return (i.e., the sum of cto and otc) is close to 3 bp per day, which corresponds to around 8 percent per annum.

3.2 Descriptive Statistics for the Main Variables

Panel B of Table 1 provides descriptive statistics for the main variables in the study. First consider our two proxies for retail attention. The average squared close-to-close return (VOL_{t-1}) is 0.127 percent, while the mean daily retail net buying ($Retail_NetBuy_{t-1}$) is not significantly different from zero. The latter result indicates that, during the 1997-2001 sample period, there was no substantive shift in ownership of Nasdaq stocks between retail and institutional investors.

Next we present three measures of the intensity of retail net buying near the day's open (NetBuy_Open, NetBuy_15Min, and NetBuy_DIFF). These mean values are significantly positive, indicating that retail investors tend to be net buyers of stocks at the open, and during the

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first fifteen minutes of the trading day. In addition, the rate of net retail buying in the first hour is significantly greater than the analogous hourly rate over the rest of the trading day.

Finally, consider our measures of short sale constraints and transaction costs. We find that mean institutional ownership (INST) is 52 percent of shares outstanding. This number is higher than the 34 percent reported in Nagel (2005), because our sample is limited to larger firms. Relative short interest averages 3.8% of shares outstanding across all stocks and days in the sample, similar to that reported in Asquith et al. (2005). The average spread at the open is 1.06 percent of the quote midpoint, while the average spread at the close is 0.62 percent, and the mean effective half spread is 0.17 percent. Finally, the average firm has a market capitalization of $5.21 Billion. Note that the medians for several variables in Panel B are smaller than their corresponding means, indicating some degree of positive skewness for these firm attributes.

3.3 Correlations across the Main Variables

In Panel C of Table 1 we report the average Spearman correlation across each pair of variables.[15] These correlations are generally consistent with our expectations. First, note that our two proxies for retail investor attention have a significant positive correlation of 6 percent. Second, both proxies are positively correlated with our measures of net retail buying pressure near the open. Third, both attention measures are positively correlated with the overnight return (cto), and negatively correlated with the trading day return (otc). All this evidence is consistent with an upwardly biased opening price for stocks that are subject to higher attention by retail investors.

Fourth, consider the association between our proxies for short sale constraints and stock returns during the overnight period versus the trading day period. The percent of institutional ownership (INST) is significantly negatively correlated with the overnight return (cto) and

15 Spearman correlations are applied to reduce the influence of outliers. Pearson correlations yield similar results.

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positively correlated with the trading day return (otc), while the opposite tendencies are apparent for relative short interest (RSI). This outcome suggests that stocks subject to more binding short sale constraints (i.e., with lower institutional ownership or higher short interest) tend to have larger overnight returns and trading day reversals. This result is also consistent with predictions based on our theory of attention-based overpricing at the open.

Fifth, the spread measures are positively correlated with the overnight return and negatively correlated with the trading day return, indicating a tendency for larger overnight returns and trading day reversals for stocks with higher transaction costs. In addition, the spread measures are negatively correlated with firm size and institutional holdings, as expected.

Finally, firm size is negatively related to overnight returns (otc), and positively related to trading day returns (cto). This outcome suggests that smaller firms have a tendency for larger positive overnight returns and trading day reversals. In addition, firm size is negatively correlated with one of our attention proxies (VOL_{t-1}), and positively correlated with institutional ownership (INST) and our second attention proxy ($Retail_Netbuy_{t-1}$). This evidence reinforces the need to control for size when we analyze the influence of institutional ownership or retail investor attention on daily returns in our analysis (see Nagel, 2005).

3.4 The Intraday Price Pattern

The descriptive statistics in Table 1 raise the question of whether this average positive overnight return and negative trading day reversal are due to a high opening price or a low closing price, or both. We address this issue by examining the pattern in prices throughout the trading day. For each stock, we collect data on intraday midquotes at 30-minute intervals. In addition, we gather the midquotes at 5-minute intervals during the first and last 30 minutes of the trading day. Then, at each 5-minute or 30-minute interval (T), we construct the ratio of the midquote at that time to

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the day's closing midquote.[16] We then find the average of each intraday price ratio across all stocks every day. Finally we compute the time series mean of these cross-sectional average price ratios for all days in the sample period.

Figure 1 plots the intraday pattern in this average ratio of the midquote at each intraday time T to the closing midquote, across all stocks and days in the sample, along with the 95% confidence interval about a ratio of one. Note that this confidence interval collapses to zero at the close, when the price ratio equals one for all firms. This intraday price pattern reveals that, on average, the opening price is significantly higher than the closing price by approximately 8 basis points (i.e., the opening price ratio = 1.0008). After the open, the price declines during the first 60 minutes of trading to within the 95% confidence interval about one, and then levels off during the middle of the day, before rising somewhat during the last few minutes of trading. This intraday pattern is consistent with the implications of our attention-based explanation, indicating that prices near the open tend to be high relative to subsequent intraday prices.[17]

4. Main Empirical Tests

4.1 Investor Attention and the Intensity of Net Retail Buying Near the Open

Our first hypothesis is that high attention days are followed by high net retail buying at the start of the next trading day. We test this hypothesis by comparing our three measures of the intensity of net retail buying near the open (NetBuy_Open, NetBuy_15Min, and NetBuy_DIFF), across portfolios partitioned by our proxies for attention, while controlling for firm size. These results appear in the first three columns of Table 2.

16 We choose the closing price to scale intraday price ratios because our main analysis focuses on open-to-close and close-to-open returns. Scaling by the average midquote between 11 a.m. and 3 p.m. yields similar results.
17 In tests not reported here, we investigate whether overnight liquidity risk can explain these overnight return patterns, in the spirit of Acharya and Pederson (2007) and Pastor and Stambaugh (2003). According to this theory, investors are willing to pay less at the daily close for stocks subject to greater overnight liquidity risk, measured by a stock's sensitivity to overnight changes in market liquidity. We find no evidence that stocks with greater overnight liquidity risk have lower closing prices, or greater overnight returns.

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In this approach we consider each trading day as a separate event. For each day, we control for firm size by initially sorting all firms into size terciles, based on the firm's mean market capitalization over the previous twenty trading days.[18] Then, within each size tercile, we form three finer portfolios by independently sorting based on each proxy for attention: VOL_{t-1} or $Retail_Netbuy_{t-1}$. Our size-adjusted high attention portfolio each day thus contains the top one third of stocks by either attention proxy, within each size tercile. We then compute the cross-sectional means of our three measures of the intensity of net retail buying near the open, for each portfolio based on low, medium, or high size-adjusted retail investor attention. Finally, we average these cross-sectional means across all days in the sample period. The t-statistics are based on the Newey-West adjusted standard errors of the time series means.

The first three columns in Panel A of Table 2 present our three measures of the average intensity of net retail buying near the open, for portfolios based on low, medium, or high attention, where the level of retail investor attention is proxied by return volatility yesterday (VOL_{t-1}). The first three columns of Panel B present the analogous results using net retail buying yesterday ($Retail_NetBuy_{t-1}$) to proxy for attention. In both Panels A and B, all three measures are significantly positive for the subsample of high attention stocks. Furthermore, the difference of means across portfolios with high versus low attention is always significantly positive at the 1% level. Together, these results confirm our first hypothesis, indicating that: (i) the intensity of retail buying near the open of the trading day is significantly greater for high attention stocks than for low attention stocks, and (ii) the intensity of retail buying of high attention stocks is significantly greater during the first hour than it is during the rest of the trading day.[19]

18 Results are robust when we do not control for size in this partitioning scheme.
19 Using net retail buy volume over the first minute or the first five minutes of the trading day yields robust results.

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4.2 Investor Attention and Overnight and Trading Day Returns

Our second hypothesis predicts that this attention-triggered retail buying pressure at the open should result in opening prices that are high relative to prices during the rest of the trading day. We test this hypothesis by comparing the overnight and trading day returns across portfolios based on low, medium, or high attention. Results appear in the next two columns of Table 2.

Panels A and B of Table 2 both reveal a strong tendency for positive mean overnight returns and negative trading day reversals, which increase in magnitude and significance as we consider stocks subject to higher levels of attention. In Panel A, when we analyze the main sample of 3,000 largest U.S. stocks, we find the mean overnight return (trading day reversal) for the high-attention portfolio is +13 bp (-13 bp) per day. In contrast, the analogous results for the subsample of low attention stocks in Panel A are much smaller, at only +3 (-3) bp per day. In Panel B, when we analyze the abridged sample of Nasdaq stocks, the analogous results for the high-attention portfolio are larger in magnitude, averaging +26 (-27) bp per day. Once again, the subsample of low-attention Nasdaq stocks in Panel B has much smaller means, at +4 (-12) bp per day. In both Panels, the mean difference t-test indicates that the overnight return (trading day reversal) is significantly larger for the portfolio of stocks subject to high retail attention.

Finally, it is noteworthy that the mean 24-hour close-to-close return in the last column of Table 2 is not significantly different from zero for any portfolio in Panel A or B. This result indicates that the significant overnight return patterns in Panels A and B represent a temporary mispricing at the open which tends to be corrected during the course of the typical trading day.

4.3 Investor Attention, Overnight and Trading Day Returns, and Institutional Ownership

In this section we consider the role of institutional ownership in concert with our proxies for retail investor attention. Stocks with low institutional ownership necessarily have a high

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concentration of retail investors, and are also difficult to short sell. For these stocks, we expect greater upward price pressure at the open on days following news that attracts the attention of retail investors (proxied by yesterday's return volatility or net retail buying). We test this hypothesis by expanding our portfolio approach to compare average overnight and trading day returns across portfolios of firms that are independently partitioned along two dimensions - the level of retail investor attention and institutional ownership - while controlling for firm size.

As before, we control for firm size by initially sorting all firms into terciles based on the firm's mean market capitalization over the previous twenty days. Then, within each size tercile, we form three finer portfolios by independently sorting based on: (1) the percentage of institutional ownership at the end of the previous quarter (INST), or (2) each proxy for retail attention yesterday (VOL_{t-1} or $Retail_NetBuy_{t-1}$). This procedure results in a 3 x 3 scheme of portfolios, sorted by size-adjusted institutional ownership and each proxy for retail attention.

Results are provided in Table 3. Panel A analyzes the main sample and uses yesterday's squared return (VOL_{t-1}) to proxy for retail investor attention at the open. This Panel presents three 3 x 3 schemes of double-sorted portfolios that provide the mean overnight returns, trading day returns, and twenty-four hour returns, respectively. On average, there are 261 stocks in each of the nine portfolios within every double-sorted partitioning scheme in Panel A.

First, consider the mean overnight returns (cto) from the left 3 x 3 scheme in Panel A of Table 3. All portfolios in this scheme have positive mean overnight returns, whose magnitudes increase with the level of retail investor attention. The bottom row of portfolios contains the largest means in this scheme, indicating that stocks with high retail attention tend to have the largest overnight returns. The mean difference t-test at the bottom of every column indicates that stocks with high retail attention significantly outperform stocks with low retail attention during

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the overnight period, conditional on institutional ownership. In addition, the left column of each scheme contains the largest element in every row, indicating that stocks with low institutional ownership tend to have the largest overnight returns. Finally, the bottom left corner cell contains stocks that are most prone to overpricing at the open, as they are subject to both high retail attention and low institutional ownership. The stocks in this cell have the largest mean overnight return, which averages +17 bp per day. In contrast, the top right corner cell contains stocks that are least prone to overpricing at the open, with low retail attention and high institutional ownership. The stocks in this cell have the smallest mean overnight return, at only +2 bp per day.

Second, consider the trading day returns (otc) from the middle 3 x 3 scheme in Panel A of Table 3. All portfolios in this scheme have negative mean trading day reversals, whose magnitudes increase with the level of retail attention. However, these mean trading day reversals are only significant for portfolios in the bottom row, among stocks with high attention, or in the left column, among stocks with low institutional ownership. Here the mean difference t-test for every column shows that stocks with high retail attention significantly underperform stocks with low attention, during the trading day. In addition, the bottom left corner cell now contains the largest negative mean trading day reversal in the scheme, at -17 bp per day, while the top right corner cell contains the smallest negative trading day reversal, at only -3 bp per day.

Third, the last 3 x 3 scheme in Panel A of Table 3 reinforces the result from the last column of Table 2, showing that attention-based overpricing at the open is a short-term intraday phenomenon. Even for the most overpriced stocks (subject to high retail attention and low institutional holdings), the significant price inflation at the open is completely reversed during the trading day, so that the mean close-to-close return is not significantly different from zero.

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Next consider the results in Panel B of Table 3, which analyze the abridged sample of Nasdaq stocks during 1997-2001, and use net retail buying yesterday to proxy for attention. These results are consistent with those in Panel A, although the price inflation at the open is substantially larger in magnitude for the abridged sample. In the first 3 x 3 scheme of Panel B, the mean overnight return for stocks most prone to overpricing at the open (i.e., the bottom left corner cell) is now +34 bp per day. This result contrasts with the top right corner cell, which contains the smallest mean overnight return, at only 2 bp per day. Similarly, in the second 3 x 3 scheme of Panel B, the mean trading day reversal for the bottom left corner cell is -42 bp, but is only -9 bp for the top right corner cell. Once again, the third 3 x 3 scheme of Panel B contains mean 24-hour returns that are smaller in magnitude and never significantly different from zero.

One likely reason for the stronger intraday price pattern for the abridged sample of Nasdaq stocks during 1997-2001, in Panel B of Table 3, is that this period represents a time of high investor sentiment (e.g., see Baker and Wurgler, 2006 and section 5 below). Furthermore, Nasdaq stocks as a group tend to have lower institutional ownership and are generally more volatile, more difficult to value, and more costly to arbitrage. We investigate these issues next.

4.4 Stocks that are Difficult to Value and Costly to Arbitrage

This section further addresses our second hypothesis, by examining the relation between overpricing at the open and variables that proxy for the difficulty in valuing a stock, and the costs of arbitrage. Both of these attributes of a stock are likely to exacerbate the extent of overpricing at the open found in Table 3 (see Baker and Wurgler, 2006, and Kumar and Lee, 2006). To this end, Table 4 reproduces the analysis in each Panel of Table 3, after restricting the samples used in Table 3 to focus on finer subsamples of stocks each day that are harder to value and more costly to arbitrage. First (1), we limit the sample to the 50% of firms each day with the highest

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return volatility over the previous 20 trading days, to represent stocks that are hard to value.[20] Second (2), we narrow this focus by considering the 50% of the hard-to-value firms in (1) with the highest mean effective half spread over the previous 20 trading days.[21] Third (3), we further confine the sample each day to the 20% of the hard-to-value, high-transaction-cost firms in (2) with the highest relative short interest during the previous month.[22]

Panel A of Table 4 presents the results after applying all three restrictions above on the main sample, using volatility yesterday (VOL_{t-1}) to proxy for retail investor attention. Panel B presents the analogous results for the abridged sample, using net retail buying yesterday ($Retail_NetBuy_{t-1}$) to proxy for attention.

First consider the results in Panel A of Table 4. After applying restrictions (1) - (3) on the main sample, there are an average of thirteen stocks within every double-sorted portfolio each day. The resulting 3 x 3 schemes in Panel A follow the same pattern as those in Table 3, although the positive overnight returns and negative trading day reversals are much larger in magnitude, as predicted. In particular, the bottom left corner cell of the first (second) 3 x 3 scheme in Panel A now reveals a mean overnight return (trading day reversal) of +43 (-45) bp.

Second, Panel B of Table 4 presents the analogous results for the abridged sample of Nasdaq stocks over the period, 1997-2001. After applying restrictions (1) - (3) on the abridged sample, there are now an average of five firms within every double-sorted portfolio each day. The resulting 3 x 3 schemes in Panel B again follow the same pattern, although the positive

20 Baker and Wurgler (2006) argue that recent volatility is a good proxy for the difficulty in valuing stocks. We have also used the firm's book-to-market ratio from the previous quarter as an alternative measure of this aspect of a stock. Results are consistent with those using return volatility over the past month, and are summarized in Table 5. We also note that stocks with high volatility are costly to sell short, due to potential margin calls. Thus, high volatility also represents a potential limit to arbitrage.

21 Sadka and Scherbina (2007) propose that high transaction costs act as an additional limit to arbitrage, and they document that such stocks underperform over time.

22 Asquith et al. (2005) argue that stocks with both low institutional ownership and high short interest are subject to more binding short sale constraints.

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overnight returns and negative trading day reversals are yet much larger. In particular, the bottom left corner cell of the first (second) 3 x 3 scheme in Panel B now reveals a mean overnight return (trading day reversal) of +61 (-70) bp per day.

Together, the results in Table 4 reinforce the view that high attention firms have positive overnight returns and trading day reversals that are larger in magnitude when the firm is more difficult to value, transaction costs are higher, and short sale constraints are more binding. This evidence provides further support for our theory of attention-based overpricing at the open.[23]

5. Variation over Time in Overpricing at the Open: The Role of Sentiment

The previous sections establish that, for a large group of stocks subject to high attention and low institutional holdings, the opening price tends to be very high relative to intraday prices. In this section we examine how the magnitude of this opening price inflation changes through time, and whether any such time-variation in this overpricing is related to changes in investor sentiment.

This analysis requires a time series measure that tracks the level of retail investor sentiment over time. We examine two alternative proxies. Our first measure is the sentiment index in Baker and Wurgler (2006, 2007). This sentiment measure is a composite index based on the first principal component of six different sentiment proxies, orthogonalized with respect to several macroeconomic conditions. The Baker-Wurgler sentiment index is available on a monthly basis from the start of our sample period to the end of 2007.[24]

Our second measure is the actual net buying activity of retail investors during the first hour of the trading day, as a percent of shares outstanding, averaged across all stocks and days

23 In tests not reported here, we have reproduced this analysis for each stage of this progression to finer subsamples of stocks that are more difficult to value and more costly to arbitrage, after applying each restriction, (1) - (3), in turn. At each stage, the 3 x 3 schemes reveal progressively larger overnight returns and trading day reversals. Results are available from the authors.

24 For details see Baker and Wurgler (2006). Their monthly sentiment index can be downloaded from Jeffrey Wurgler's website at www.stern.nyu.edu/~jwurgler.

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during the month. This measure is motivated by the results in Table 2, which indicate that net retail buying near the day's open is an effective proxy for cross-sectional differences in retail investor sentiment based on attention. We refer to this measure as our retail sentiment index. The available data on net retail buying enable measurement of this sentiment index on a monthly basis for the abridged sample of Nasdaq stocks over the period, 1997-2001.

Our tests in this section are designed to establish whether monthly variation in these two proxies for retail investor sentiment can explain variation through time in the extent to which opening prices are pushed above 'fundamental' values (i.e., values during the rest of the trading day). In order to measure the extent of opening price inflation during a given month, we first calculate daily overnight and trading day returns for groups of stocks classified using the same procedures as before. That is, for each day we compute the mean overnight and trading day returns for the 3 x 3 scheme of portfolios sorted along two dimensions: size-adjusted institutional ownership and retail investor attention yesterday. Next, for each month, we calculate the mean daily overnight and trading day returns to measure the average daily performance of each portfolio during the overnight and trading day periods, respectively, throughout that month. Thus our monthly overnight (trading day) return measure for each portfolio, CTO_m (OTC_m), contains the mean of daily overnight (trading day) returns during each month (m).

We then examine the monthly performance of a zero-cost hedge portfolio that is long the stocks that are most subject to inflation of opening prices (i.e., the bottom left corner portfolio in the 3 x 3 scheme), and short the stocks that are least subject to inflation of opening prices (i.e., the top right corner portfolio). That is, each day this hedge portfolio is long stocks with low institutional holdings and high attention (bottom left corner), and short stocks with high institutional ownership and low attention (top right corner). The monthly mean trading day

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(open-to-close) return on this hedge portfolio represents the extent to which the opening price exceeds the close, for stocks most prone to opening price inflation relative to stocks least prone to this overpricing. Likewise, this monthly average trading day return on the hedge portfolio provides one measure of changes over time in market-wide hidden costs associated with the tendency for retail investors to buy attention-grabbing stocks at the open (rather than the close).

Figure 2 plots the evolution over the period, 1996-2007, of several variables that measure retail investor sentiment, the extent of overpricing at the open, and transaction costs. First, we plot in Figure 2 movements in the Baker-Wurgler sentiment index over this period. Second, we also plot the average return on the hedge portfolio during the overnight and trading day periods, respectively, using yesterday's return squared as the attention proxy. Finally, we plot the daily average (negative value of the) effective half spread for the subsample of stocks most sensitive to price inflation at the open (i.e., stocks with low institutional ownership and high attention).[25]

Figure 3 presents the analogous information using our retail sentiment index, for the abridged sample of Nasdaq stocks over the period, 1997-2001. In this figure we compile the returns on the hedge portfolio using yesterday's net retail buying as our proxy for attention. For comparison, we also plot the Baker-Wurgler sentiment index and the (negative value of the) mean effective half spread for the portfolio of stocks most subject to opening price inflation.[26]

First consider the information in Figure 2. This plot reveals that overnight returns on the hedge portfolio are positively related to the Baker-Wurgler sentiment index (correlation is .30, p-value is .001), while trading day returns are negatively related to this index (correlation is -.33, p-value is .001). Similarly, we find that the retail sentiment index in Figure 3 has a correlation of .47 (p-value is .001) with overnight returns on the hedge portfolio, and a correlation of -.51 (p-

25 For clarity, we divide the Baker-Wurgler sentiment index by 2 and report quarterly averages of all variables.
26 Here we divide the Baker-Wurgler sentiment index by 2, multiply the retail sentiment index with 10, and report quarterly averages of the variables.

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value is .001) with trading day returns on the hedge portfolio. Furthermore, in Figure 3 the Baker-Wurgler index is highly correlated with the retail sentiment index. Based on the 60 months for which we have data on both indices, we find a correlation of .77 (p-value is .001).

It is noteworthy that the positive overnight returns on the hedge portfolio in Figures 2 and 3 become very large just prior to the technology crash in March of 2000, and remain positive throughout 2001. In comparison, the negative trading day reversals on the hedge portfolio become even larger as this crash develops. Apparently, the bulk of the 2000 correction in stocks most subject to opening price inflation occurred during the trading day.[27]

Figures 2 and 3 also plot the average effective half spread for the subsample of stocks most prone to price inflation at the open (i.e., those subject to high attention and low institutional ownership). For comparison purposes, the negative value of this measure of transaction costs is plotted alongside the trading day (open-to-close) return of the hedge portfolio. Once again, we regard this trading day return on the hedge portfolio as a measure of the implicit cost of buying at the open (rather than the close) those stocks most prone to opening price inflation. Both Figures show that this implicit cost of buying at the open is similar in magnitude to the effective half spread, for much of the sample period. Moreover, in times of high investor sentiment, this mean trading day reversal can be more than twice the size of the effective half spread.

The major practical implication of this analysis is that retail traders incur a substantial hidden transaction cost when they buy attention-grabbing stocks at the open of the next trading day, because their herding behavior results in inflated opening prices. The plots reveal that, over long periods, this implicit transaction cost is similar in magnitude to the effective half spread.

27 We also note that the Baker-Wurgler sentiment index in Figures 2 and 3 peaks a few months after March of 2000, and after the peak of our Retail Investor sentiment index in Figure 3. This delay in the peak of the Baker-Wurgler index may be due to their use of a 12-month lag for three of the six components in their index. They argue that combining these lagged values for three of their series with current values of the other three series best captures annual variation in the six series (see Baker and Wurgler, 2006, 2007).

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6. Robustness Tests

6.1 Alternative Portfolio Approaches

In Panel A of Table 5, we apply the main analysis from our portfolio approach of Table 3, but we base this analysis on alternative return measures, or various subsamples of stocks, or different time periods. For each test, the four columns on the left side of Table 5 report results based on the main sample of 3,000 largest U.S. stocks over the period, 1996-2008, where the level of retail investor attention is proxied by return volatility yesterday (VOL_{t-1}). The right four columns present the analogous results based on the abridged sample of Nasdaq stocks over the period, 1997-2001, using retail net buying yesterday ($Retail_NetBuy_{t-1}$) to proxy for attention.

For brevity, in all of these robustness tests we only present the mean returns for the two extreme corner portfolios of every 3 x 3 scheme, for stocks with low institutional holdings and high attention (bottom left corner) and for stocks with high institutional holdings and low attention (top right corner). The base-case provided at the top of Table 3 reproduces the results for these two corner portfolios from Table 3. In every subsequent robustness test (row) of Panel A, we change only one aspect of the analysis to facilitate comparison with the base case.

The series of tests in Panel A reveals that our results are robust when we use market-adjusted returns, median returns, or trade prices to measure returns, and when we exclude low-price stocks. This behavior is also ubiquitous across Nasdaq and NYSE stocks, small and large stocks, growth and value stocks, and momentum winners and losers, although there is a larger mean overnight return and trading day reversal for small stocks, Nasdaq stocks, growth stocks, and momentum losers. We also find these patterns are larger in magnitude on Mondays, but they are also present on the other days of the week. In addition, these results appear in all subperiods, though they decline in magnitude over time, perhaps due to regulatory changes and technological advances, such as decimalization in 2001 and the subsequent expansion of algorithmic trading.

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6.2 Fama-MacBeth Regression Analysis

In this section we examine whether our results are robust to alternative methodology, by applying regression analysis based on the following model:[28]

$$cto_{it}\ (otc_{it}) = a_0 + a_1 ATTENTION_{it} + a_2 INST_{it} + a_3 SIZE_{it} + \varepsilon_{it}, \qquad (1)$$

where $ATTENTION_{it}$ = each proxy for retail investor attention (VOL_{t-1} or $Retail_NetBuy_{t-1}$);
$INST_{it}$ = the firm's percentage institutional ownership from the previous quarter;
$SIZE_{it}$ = the firm's mean market capitalization over the previous twenty trading days.

In this analysis, the dependent variables are winzorized to eliminate outliers beyond one percent in the tails. In addition, the independent variables are replaced with their decile ranks, adjusted to range between -0.5 and +0.5. This rank regression approach attenuates the influence of outliers.[29]

First, for every day (t) in the sample period, we estimate model (1) for the cross-section of stocks (i), using each dependent variable (cto or otc). Second, we compute the time series mean coefficients across all days (t) in the sample period. The standard errors of these Fama-MacBeth mean coefficients are used to construct the t-ratios.

Results appear in Panel B of Table 5. They show that the overnight return is positively related to each proxy for retail investor attention yesterday, and negatively related to the firm's institutional ownership. Likewise, the trading day reversal is negatively associated with each attention proxy, and positively related to institutional ownership. Consistent with our theory of attention-based overpricing at the open, these results imply a larger overnight return and trading day reversal for stocks that are subject to higher levels of retail investor attention yesterday, along with greater limits to arbitrage embodied in lower institutional ownership. These results also reveal that smaller firms tend to have a larger trading day reversal. We conclude that our evidence and its implications are not dependent upon the research methodology applied.

28 We have also included the firm's effective half spread and relative short interest in model (1), with robust results.
29 Similar results are obtained when the analysis is applied to the original data.

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7. Summary and Conclusions

This paper investigates the role of attention as a potential source of retail investor sentiment, and examines whether attention-based retail trading causes prices to temporarily deviate from fundamental values at the open of the typical trading day. We show that high attention days for individual stocks are followed by high net retail buying at the start of the next trading day. Moreover, this increased retail buying pressure results in opening prices that tend to be high relative to prices during the rest of the trading day. Furthermore, we find that these predictable intraday price patterns are more pronounced for stocks that are difficult to value and costly to arbitrage, and they are larger in magnitude during periods of high investor sentiment.

This study is the first to examine intraday patterns in retail trading and price formation in relation to investor sentiment, proxied by retail investor attention. By investigating our theory of attention-based overpricing at the open, we uncover and help resolve a here-to-fore unexplained intraday price anomaly: stocks that have recently attracted the attention of retail investors have a strong tendency for net retail buying at the open, which leads to high opening prices and positive returns during the overnight period followed by reversals during the subsequent trading day.

Our results should be of interest to investors and regulators alike. During periods of high investor sentiment, a strategy of selling high attention stocks at the open and delaying purchases until later in the trading day can yield significant improvements to performance. This evidence shows that some retail investors bear substantive hidden costs, through their penchant for buying stocks at inflated prices during the first hour of trading following high attention days. During periods of high investor sentiment, for example, the implicit costs of buying high attention stocks at the start of the day can be twice the effective half spread. The first step towards reducing this hidden cost of buying high attention stocks at the open is to enhance awareness among retail traders of the potentially detrimental price impact of such trading activity.

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Figure 1. Intraday Price Pattern
Ratio of Midquote at Intraday Time (T) to the Closing Midquote across the 3,000 Largest U.S. Stocks and all Days: 1996 - 2008

This intraday price pattern traces out the ratio of the midquote at different times (T) during the trading day, to the closing midquote. These intraday price ratios are computed for each stock at 5-minute intervals over the first and last 30-minutes of the trading day, and at 30-minute intervals over the rest of the trading day. First, for each day we average this price ratio at every intraday time (T) across all stocks in the sample. Second, we compute the time series mean of these daily cross-sectional averages across all days in the sample period. The 95 percent confidence interval about a ratio of 1.0 is constructed using the standard error of the time series mean across all days, at each time (T).

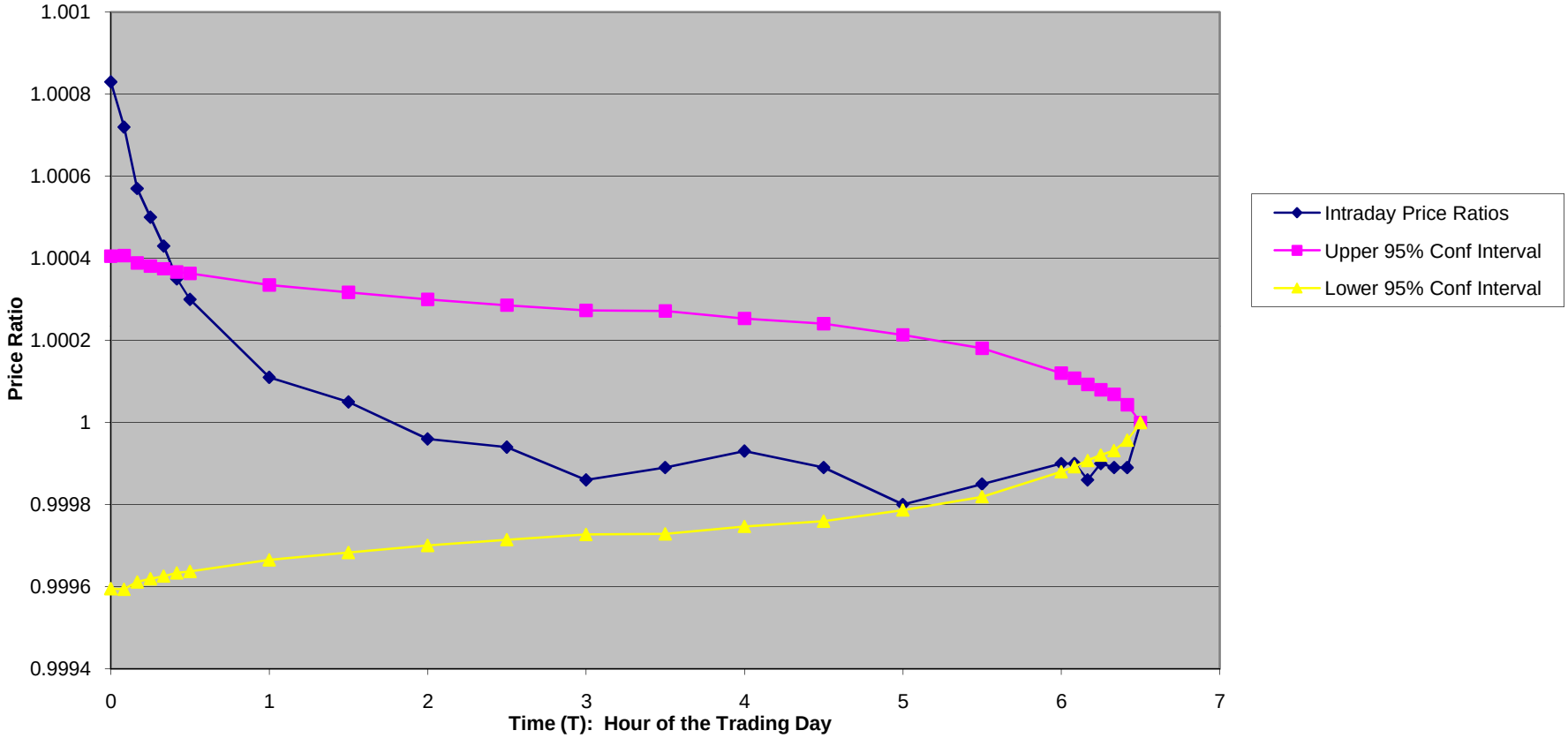
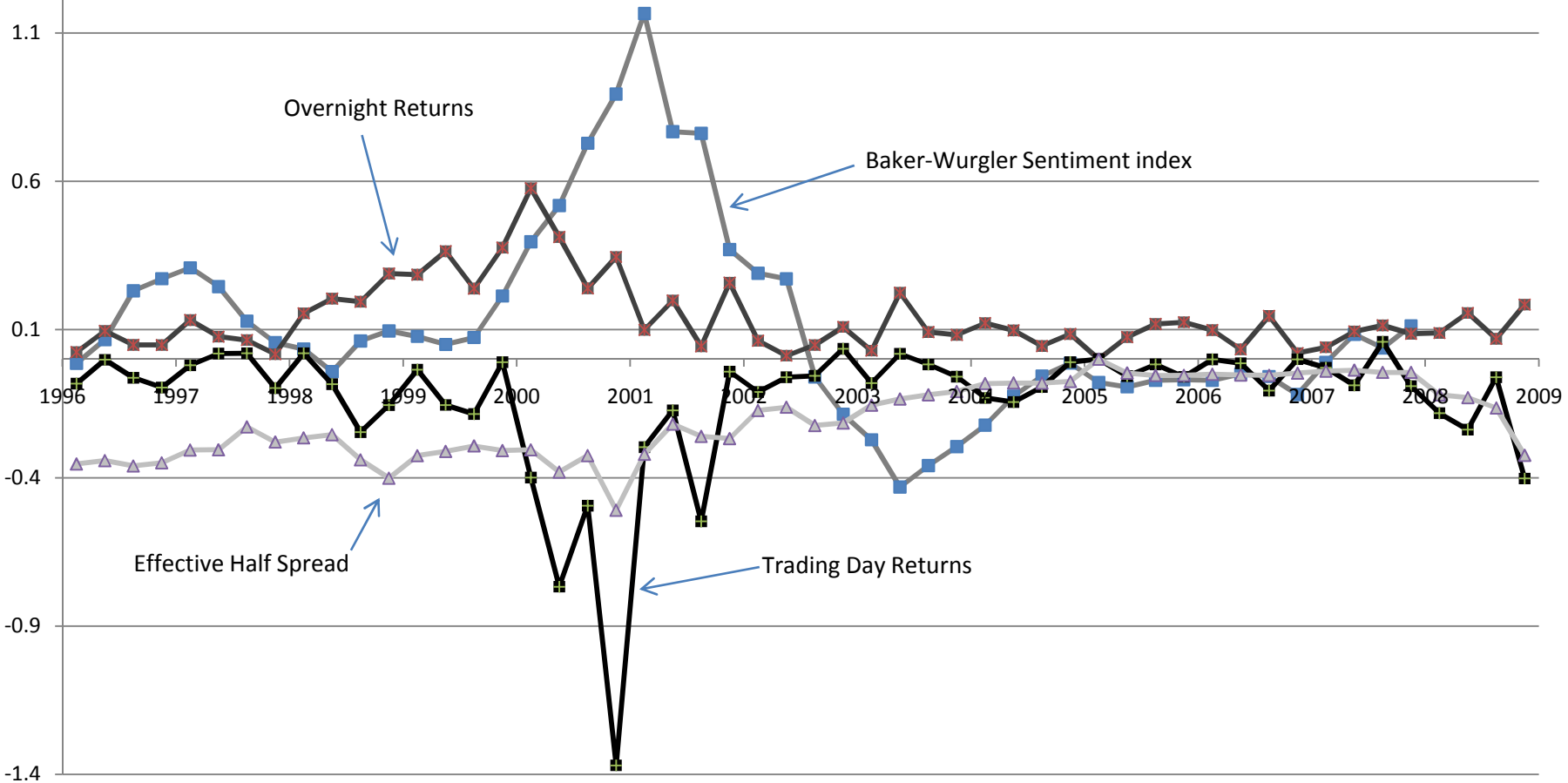


Figure 2. Retail Investor Sentiment, Overnight Returns, and Transaction Costs: 1996 - 2008

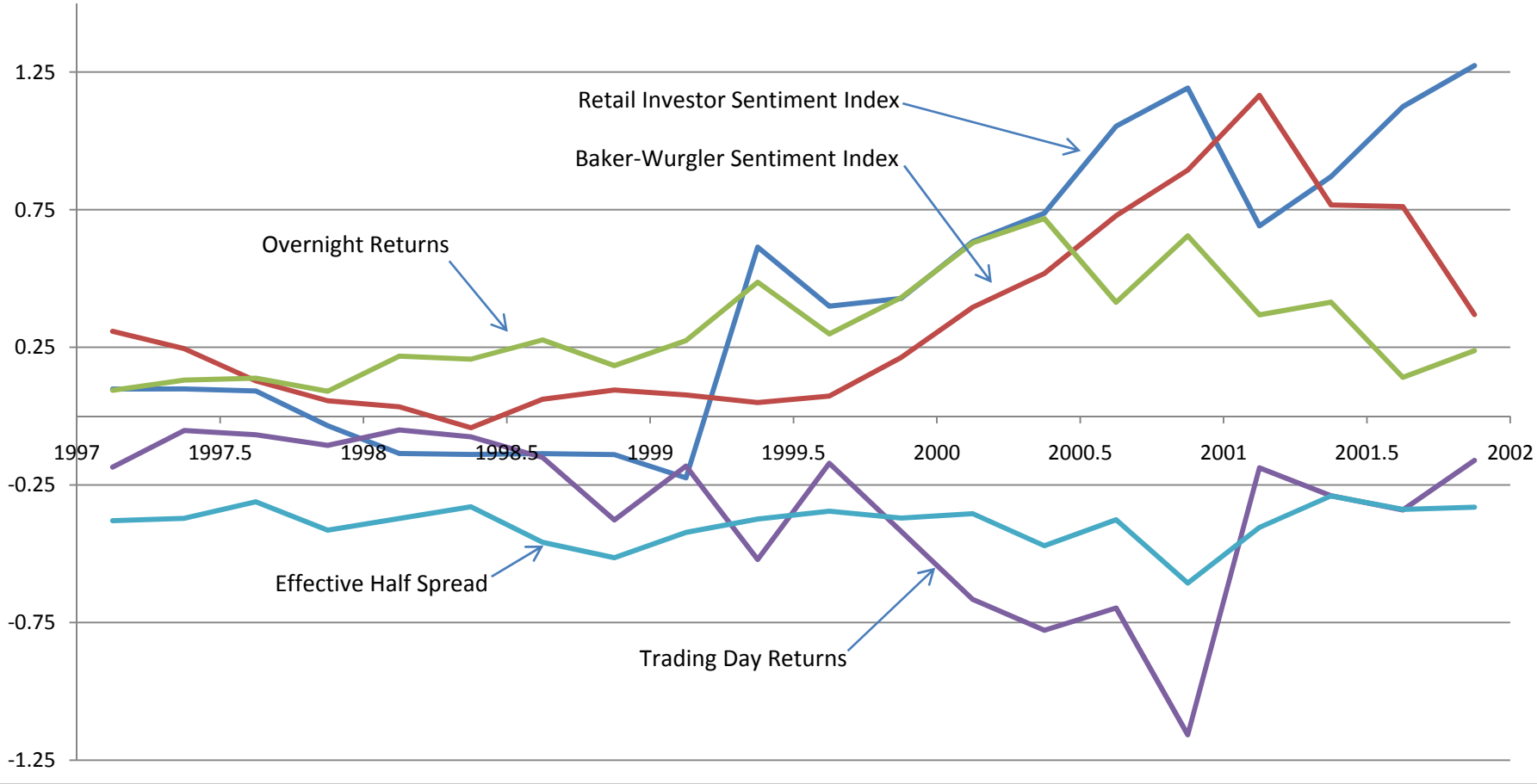
This Figure presents the evolution over the period, 1996 - 2007, of several variables that measure retail investor sentiment, the extent of overpricing at the open, and transaction costs. First, we report quarterly averages of the monthly Baker-Wurgler sentiment index. Next, for each quarter, "Overnight Returns" plots the average daily overnight (close-to-open) return of a zero-cost hedge portfolio that is long stocks with low institutional holdings and high attention (i.e., the bottom left corner of the 3 x 3 scheme in Table 3), and short stocks with high institutional ownership and low attention (i.e., top right corner of the scheme). "Trading Day Returns" plots the average trading day (open-to-close) return on the same hedge portfolio. We also plot the average daily effective half spread for the subsample of stocks that are most sensitive to price inflation at the open (i.e., stocks in the bottom left corner of the scheme, with high attention and low institutional ownership).



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Figure 3. Retail Investor Sentiment, Overnight Returns, and Transaction Costs: 1997 - 2001

This Figure presents the evolution over the period, 1997 - 2001, of several variables that measure retail investor sentiment, the extent of overpricing at the open, and transaction costs. We report quarterly averages of the monthly Baker-Wurgler sentiment index, and the quartely average of the Retail Investor Sentiment Index (defined as the actual net buying activity by retail investors during the first hour of the trading day, as a percent of shares outstanding, averaged across all stocks and all days during the quarter). For each quarter, "Overnight Return" plots the average daily overnight (close-to-open) return of a zero-cost hedge portfolio that is long stocks with low institutional holdings and high attention (bottom left corner of the 3 x 3 scheme), and short stocks with high institutional ownership and low attention (top right corner of the scheme). "Trading Day Return" plots the average trading day (open-to-close) return on the same hedge portfolio. We also plot the average daily effective half spread, for the subsample of stocks that are most sensitive to price inflation at the open (i.e., stocks in the bottom left corner of the scheme, with high attention and low institutional ownership).



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Table 1. Descriptive Statistics and Correlations across Variables

We apply our analysis to two major samples. Our main analysis is applied to the subsample of the 3,000 largest U.S. stocks each year with nonmissing TAQ data on quotes, over the 13-year period covering 1996 - 2008. We also analyze an abridged sample of all Nasdaq firms for which data on intraday retail trading activity are available, over the period covering 1997 - 2001. The focus of this study is on percentage overnight and trading day returns, measured as 100 times the log of the price relative using quote midpoints at the open and the close: cto = close-to-open (overnight) return, and otc = open-to-close (trading day) return. We also analyze the difference between overnight and trading day returns (DIFF = cto - otc), as well as this same difference after deducting transaction costs measured as the average spread at the day's open and close: DIFF-TC = DIFF - [SPR(open)+SPR(close)] / 2. In addition, we examine daily returns measured over the 24-hour periods from close-to-close and from open-to-open: ctc and oto.

We consider two alternative proxies for retail investor attention: (i) VOL_{t-1} = the squared 24-hour (close-to-close) return yesterday, and (ii) $Retail_NetBuy_{t-1}$ = the net number of shares bought by retail investors yesterday, as a percent of total share volume.

Next, we compute three measures of the intensity of net retail buying near the open:

NetBuy_open = the mean proportion of stocks each day for which the first trade of the day is a purchase by a Retail Trader, minus 0.5;
NetBuy_15Min = (Net Retail Buy Volume) / (Total Retail Share Volume), over the first 15 minutes of the trading day;
NetBuy_Diff = [(Net Retail Buy Volume in 1st Hour) - (Net Retail Buy Volume in Last 5.5 Hours / 5.5)] / (shares outstanding).

The third measure defined above represents the difference between the hourly rate of net buying by retail investors during the first hour of trading, and the analogous hourly rate during the rest of the trading day, as a percent of shares outstanding.

We also consider two proxies for short sale constraints and three measures of transaction costs. Our two proxies for short sale constraints are: (i) the percent of institutional ownership (INST), and (ii) relative short interest (RSI) measured as the number of shares sold short each month divided by shares outstanding. Our three measures of daily transaction costs include: (i) the bid-ask spread as a percent of the quote midpoint at the open (SPR(open)), (ii) the analogous percent spread at the close (SPR(close)), and (iii) the daily effective half spread (Spread). Finally, firm size (SIZE) is proxied by daily market capitalization.

The descriptive statistics in Panels A and B are calculated by first computing the cross-sectional mean (or median) each day, and then averaging these means (or medians) across all days in the sample period. The standard deviation of the time series average across daily means is then used to construct the t-test for each statistic in Panels A and B. Similarly, the Spearman correlations in Panel C are calculated by first computing the cross-sectional correlation each day, and then averaging these correlations across all days in the sample. Once again, the standard deviation of the time series average correlation is used to construct the t-test for each average correlation in Panel C.

Panel A: Descriptive Statistics for Overnight (Close-to-Open) Returns, Trading Day (Open-to-Close) Returns, and 24-Hour Returns

	Overnight & Trading Day Returns				24-Hour Returns	
	cto	otc	DIFF	DIFF - TC	ctc	oto
Mean (%)	.098	-.066	.165	-.708	.031	.028
Median (%)	.045	-.061	.108	-.618	-.016	.-.027
T (H_0: Mean = 0)	5.2 **	-3.3 **	6.1 **	-24.8 **	1.1	1.2
Avg # of Firms per Day	2,447	2,460	2,447	2,383	2,447	2,604

* indicates significance at the .05 level; and ** indicates significance at the .01 level.

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Table 1, continued

Panel B: Descriptive Statistics for Attention, Net Retail Buying Near the Day's Open, Short Sale Constraints, Transaction Costs, and Firm Size

	Proxies for Attention		Net Retail Buying Near the Open			Short Sale Constraints		Transaction Costs			Firm Size
	VOL_{t-1}	$Retail_NetBuy_{t-1}$	First Trade NetBuy_Open	First 15-Min NetBuy_15Min	(First Hour - Rest) NetBuy_DIFF	INST	RSI	SPR(open)	SPR(close)	Spread	SIZE ($000)
Mean (%)	.127	.000	.004	.024	.025	.520	.038	1.06	.62	.17	5,210,818
Median (%)	.059	.136	.052	.038	.004	.532	.022	.72	.41	.12	1,109,367
T (H_0: Mean=0)	31.3 **	-.3	3.4 **	15.4 **	26.2 **	214.6 **	101.1 **	83.6 **	68.3 **	81.1 **	175.8 **
Avg # Firms/Day	2,602	401	161	310	400	2,681	2,681	2,616	2,619	2,463	2,534

Panel C: Spearman Correlations

VOL_{t-1}	1.00										
$Retail_NetBuy_{t-1}$	.062 **	1.00									
NetBuy_Open	.002	.038 **	1.00								
NetBuy_15Min	.004 **	.085 **	.465 **	1.00							
NetBuy_DIFF	.027 **	.051 **	.135 **	.369 **	1.00						
INST	.082 **	.004 **	.006 **	.010 **	.025 **	1.00					
RSI	.358 **	.044 **	.006 **	.003	.034 **	.390 **	1.00				
SPR(open)	.173 **	-.027 **	-.006 **	-.009 **	-.043 **	-.136 **	-.073 **	1.00			
SPR(close)	.158 **	-.025 **	-.005 **	-.007 **	-.044 **	-.189 **	-.126 **	.465 **	1.00		
Spread	.332 **	-.021 **	-.001	-.008 **	-.052 **	-.283 **	-.084 **	.569 **	.570 **	1.00	
SIZE	-.223 **	.030 **	.001	.009 **	.051 **	.163 **	.008	-.563 **	-.531 **	-.705 **	1.00
cto	.055 **	.058 **	.038 **	.028 **	.016 **	-.011 **	.020 **	.021 **	.006 **	.020 **	-.017 **
otc	-.044 **	-.020 **	.000	-.006 **	.051 **	.009 **	-.023 **	-.005 **	-.018 **	-.024 **	.029 **
DIFF	.056 **	.037 **	.011 **	.011 **	-.045 **	-.011 **	.027 **	.009 **	.017 **	.025 **	-.028 **
DIFF - TC	.000	.037 **	.010 **	.011 **	-.040 **	.040 **	.061 **	-.203 **	-.166 **	-.134 **	.119 **

* indicates significance at the .05 level; and ** indicates significance at the .01 level.

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Table 2. Net Retail Buying Near the Open of the Trading Day, Overnight Returns, Trading Day Returns, and 24-Hour Returns, Across Portfolios Sorted by our Two Proxies for Retail Investor Attention: Return Volatility and Net Retail Buying Yesterday

In columns (1) and (2), this table presents information about the intensity of net retail buying at the open, and during the first fifteen minutes of the trading day:
(1) NetBuy_Open = the mean proportion of stocks each day for which the first trade of the day is a purchase by a Retail Trader, minus 0.5.
(2) NetBuy_15Min = (Net Retail Buy Volume) / (Total Retail Share Volume) over the first 15 minutes of the trading day.
Next, in column (3) we report the difference between the hourly rate of net buying by retail investors during the first hour of trading, and the analogous hourly rate during the rest of the trading day, as a percent of shares outstanding. This measure is defined as follows:
(3) NetBuy_DIFF = [(Net Retail Buy Volume in 1st Hour) - (Net Retail Buy Volume in Last 5.5 Hours / 5.5)] / (shares outstanding).
In columns (4) - (6), we give the mean returns over three time frames: overnight (close-to-open), trading day (open-to-close), and 24 hours (close-to-close).
We report all of these measures for a 3 x 1 scheme of portfolios where stocks are partitioned each day into terciles by the level of retail investor attention. For every portfolio, we first compute the mean value of each measure across firms every day, and then average these cross-sectional means across all days in the sample.

Panel A. Using Return Volatility Yesterday (VOL_{t-1}) to Proxy for Retail Investor Attention: 3,000 Largest U.S. Stocks, 1997 - 2001

	(1)	(2)	(3)	(4)	(5)	(6)
Attention	NetBuy_Open[a] First Trade	NetBuy_15Min[a] First 15-Min	NetBuy_DIFF[a] (First Hour - Rest)	Overnight cto	Trading Day otc	24-Hour ctc
Low	-.003	.018 **	.014	.03 **	-.03	.00
Medium	.002	.022 **	.021	.04 **	-.05 *	.00
High	.012 **	.030 **	.045 **	.13 **	-.13 **	.01
High - Low	.015	.012	.031	.11	-.10	.01
T-stat	6.2 **	4.3 **	10.9 **	7.7 **	-3.1 **	0.1

Panel B. Using Retail Net Buying Yesterday ($Retail_NetBuy_{t-1}$) to Proxy for Retail Investor Attention: Nasdaq Stocks, 1997 - 2001

Attention	NetBuy_open First Trade	NetBuy_15Min First 15-Min	NetBuy_Diff (First Hour - Rest)	Overnight cto	Trading Day otc	24-Hour ctc
Low	-.017 **	-.034 **	-.004 **	.04 *	-.12 *	-.08
Medium	.004 *	.024 **	.020 **	.16 **	-.16 **	.00
High	.024 **	.079 **	.060 **	.26 **	-.27 **	.00
High - Low	.041	.113	.064	.22	-.15	.07
T-stat	18.8 **	37.4 **	24.3 **	6.8 **	-2.1 *	0.7

[a] These measures of net retail buying near the open are computed for all Nasdaq Stocks, 1997 - 2001. * indicates significance at the .05 level; ** at the .01 level.

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Table 3. Overnight Returns, Trading Day Returns, and 24-Hour Returns across Portfolios Double-Sorted by Proxies for Retail Investor Attention Yesterday, and Institutional Ownership

This table reports mean overnight returns (close-to-open or cto), trading day returns (open-to-close or otc), and 24-hour returns (close-to-close or ctc), for a 3 x 3 scheme of portfolios partitioned each day into terciles along two dimensions. The portfolios in each row are sorted according to a low, medium, or high level of retail investor attention yesterday; the columns are sorted by institutional ownership in the previous quarter.

In Panel A we proxy retail investor attention using stock return volatility lagged one day. In Panel B we use the net number of shares bought by retail investors all day yesterday, as a percent of total share volume. For every portfolio in each 3 x 3 scheme, we first compute the mean return across firms every day, and then average these cross-sectional means across all days in the sample period.

Panel A. Using Stock Return Volatility Yesterday to Proxy for Retail Investor Attention: 3,000 Largest U.S. Stocks, 1996-2008

Retail Investor Attention	Overnight Return (cto%)			Trading Day Return (otc%)			24-Hour Return (ctc%)		
Stock Return Volatility	Institutional Ownership			Institutional Ownership			Institutional Ownership		
Yesterday	Low	Medium	High	Low	Medium	High	Low	Medium	High
Low	.03 **	.02 **	.02 **	-.03 *	-.03	-.03	.00	.00	.00
Medium	.05 **	.04 **	.04 **	-.06 **	-.04	-.04	-.01	.01	.00
High	.17 **	.13 **	.10 **	-.17 **	-.12 **	-.09 **	.00	.01	.01
High - Low	.13	.11	.08	-.13	-.09	-.07	.00	.02	.01
T-stat	8.9 **	7.7 **	5.8 **	-4.0 **	-2.7 **	-2.0 *	0.0	0.4	0.3

[a] On average, there are 261 firms in the portfolio for each cell of these 3 x 3 partitioning schemes. * indicates statistical significance at the .05 level; ** at the .01 level.

Panel B. Using Net Retail Buying Yesterday to Proxy for Retail Investor Attention: Nasdaq Stocks, 1997-2001

Retail Investor Attention	Overnight Return (cto%)			Trading Day Return (otc%)			24-Hour Return (ctc%)		
Net Retail Buying	Institutional Ownership			Institutional Ownership			Institutional Ownership		
Yesterday	Low	Medium	High	Low	Medium	High	Low	Medium	High
Low	.09 **	.06 *	.02	-.19 **	-.15 **	-.09	-.09	-.09	-.07
Medium	.23 **	.18 **	.13 **	-.28 **	-.21 **	-.10	-.06	-.03	.02
High	.34 **	.31 **	.24 **	-.42 **	-.30 **	-.22 **	-.08	.01	.02
High - Low	.24	.25	.22	-.23	-.15	-.13	.01	.10	.09
T-stat	6.4 **	7.1 **	6.5 **	-2.9 **	-2.0 *	-1.7	0.1	1.1	1.0

[a] On average, there are 90 firms in the portfolio for each cell of these 3 x 3 partitioning schemes. * indicates statistical significance at the .05 level; ** at the .01 level.

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Table 4. Overnight Returns, Trading Day Returns, and 24-Hour Returns Across Double-Sorted Portfolios, for Subsamples of Stocks Each Day: (1) that are Hard to Value, (2) with High Transaction Costs, and (3) with High Short Interest

This Table follows the same format as Table 3. Panel A uses yesterday's return volatility to proxy for retail investor attention, in order to analyze the subsample of stocks each day: 1. that are Hard-to-Value (i.e., the 50% of stocks with the highest return volatility over the past 20 days), 2. with High Transaction Costs (i.e., the 50% of stocks in 1 with the highest effective half spread), and 3. with High Short Interest (i.e., the 20% of stocks in 2 with the highest relative short interest). Panel B provides analogous results using yesterday's net retail buying to proxy for retail attention.

Panel A. Using Stock Return Volatility Yesterday to Proxy for Retail InvestorAttention: Largest 3,000 U.S. Stocks, 1996-2008

Retail Investor Attention	Overnight Return (cto%)			Trading Day Return (otc%)			24-Hour Return (ctc%)		
Stock Return Volatility	Institutional Ownership			Institutional Ownership			Institutional Ownership		
Yesterday	Low	Medium	High	Low	Medium	High	Low	Medium	High
Low	.15 **	.07	.12 **	-.33 **	-.27 **	-.18 **	-.19 **	-.20 **	-.07
Medium	.22 **	.19 **	.14 **	-.35 **	-.24 **	-.19 **	-.13 **	-.06	-.05
High	.43 **	.39 **	.29 **	-.45 **	-.35 **	-.30 **	-.02	.04	-.01
High - Low	.28	.32	.18	-.12	-.08	-.12	.16	.24	.06
T-stat	8.6 **	6.4 **	6.0 **	-1.8	-1.1	-1.8	2.2 *	2.7 **	0.8

[a] On average, there are 13 firms in the portfolio for each cell of these 3 x 3 partitioning schemes. * indicates statistical significance at the .05 level; ** at the .01 level.

Panel B. Using Net Retail Buying Yesterday to Proxy for Retail Investor Attention: Nasdaq Stocks, 1997-2001

Retail Investor Attention	Overnight Return (cto%)			Trading Day Return (otc%)			24-Hour Return (ctc%)		
Net Retail Buying	Institutional Ownership			Institutional Ownership			Institutional Ownership		
Yesterday	Low	Medium	High	Low	Medium	High	Low	Medium	High
Low	.35 **	.27 **	.10	-.43 **	-.47 **	-.24 *	-.08	-.20	-.14
Medium	.47 **	.46 **	.27 **	-.71 **	-.61 **	-.43 **	-.24	-.15	-.15
High	.61 **	.65 **	.49 **	-.70 **	-.67 **	-.52 **	-.09	-.02	-.03
High - Low	.26	.38	.39	-.27	-.20	-.28	-.01	.18	.11
T-stat	3.3 **	4.8 **	5.0 **	-1.8	-1.2	-1.9	-0.1	1.0	0.7

[a] On average, there are 5 firms in the portfolio for each cell of these 3 x 3 partitioning schemes. * indicates statistical significance at the .05 level; ** at the .01 level.

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Table 5. Robustness Tests: Alternative Portfolio Approaches and Regression Analysis

For every robustness test in Panel A, we reproduce the portfolio approach from Table 3, but we change some aspect of the analysis. For brevity, we present only the two cells in each 3 x 3 scheme that are subject to the greatest and least forces behind overpricing at the open: the bottom left corner cell and the top right corner cell. The base-case reproduces these two corner cells from the 3 x 3 schemes in Table 3. In each subsequent test (row) of Panel A, we change only one aspect of the analysis to facilitate comparison with the base case.

Panel B provides the results of Fama-MacBeth regression analysis. First, for each day in the sample period we estimate the cross-sectional regression model in equation (1), for overnight returns (cto) or for trading day returns (otc). Second, we compute the time series mean coefficients across all days in the sample period.

Panel A. Portfolio Approach: Alternative Return Measures and Subsamples

Bottom Left and Top Right Corner Cells in Each 3x3 Matrix

Sample:	3,000 U.S. Stocks, 1996-2008				Nasdaq Stocks, 1997-2001			
Attention Proxy:	VOL_{t-1}				$Retail_NetBuy_{t-1}$			
Return Measure:	cto(%)		otc(%)		cto(%)		otc(%)	
Cell in 3x3 Matrix :	Bottom Left	Top Right	Bottom Left	Top Right	Bottom Left	Top Right	Bottom Left	Top Right
Base Case (Table 3)	.17	.02	-.17	-.03	.34	.02	-.42	-.09
Newey-West t-ratio	11.9 **	3.5 **	-5.6 **	-1.3	11.3 **	1.0	-6.8 **	-1.8
1. Using Market-Adjusted Abnormal Returns	.12	-.03	-.18	-.03	.24	-.07	-.42	-.09
	14.1 **	-4.2 **	-10.1 **	-3.5 **	12.3 **	-5.6 **	-9.8 **	-3.0 **
2. Using Trade Prices[a] to Measure Returns	.17	.02	-.17	-.01	.33	.00	-.41	-.08
	10.1 **	2.7 **	-4.8 **	-.3	11.0 **	.2	-6.8 **	-1.4
3. Using Median Return across Stocks Each Day	.09	.02	-.16	-.04	.20	.01	-.42	-.11
	9.2 **	3.4 **	-6.5 **	-2.5 *	9.5 **	.8	-8.7 **	-2.5 *
4. No Low-Price Stocks (with Price < $5)	.13	.02	-.13	-.02	.30	.02	-.37	-.09
	9.4 **	2.9 **	-4.5 **	-1.1	10.0 **	.8	-6.0 **	-1.7
5. Nasdaq Stocks	.28	.03	-.27	-.07	.34	.02	-.42	-.09
	13.9 **	3.6 **	-7.1 **	-2.5 *	11.3 **	1.0	-6.8 **	-1.8
6. NYSE Stocks	.09	.01	-.07	-.01	–	–	–	–
	6.0 **	.8	-3.1 **	-.7	–	–	–	–
7. Small Stock Quintile[b]	.29	.07	-.19	-.05	.45	.07	-.39	-.08
	18.3 **	8.6 **	-5.3 **	-2.1 *	13.8 **	3.3 **	-5.9 **	-1.4
8. Large Stock Quintile[b]	.09	.01	-.11	.00	.32	.07	-.37	-.21
	5.1 **	1.5	-3.9 **	.2	6.8 **	1.9	-4.2 **	-2.9 **
9. Growth Stocks (Low Quintile, B / M)[c]	.20	.03	-.23	-.04	.37	.04	-.46	-.14
	9.8 **	3.4 **	-6.0 **	-1.5	8.6 **	1.2	-5.5 **	-1.9
10. Value Stocks (High Quintile, B / M)[c]	.13	.04	-.09	-.03	.23	-.01	-.24	-.01
	9.5 **	4.7 **	-3.0 **	-1.5	6.2 **	-.3	-3.5 **	-.2

* indicates statistical significance at the .05 level, and ** indicates significance at the .01 level.

[a] Results based on trade prices using VOL_{t-1} are computed over the sample period, 1996 - 2004.

[b] For the Nasdaq stocks in the right four columns, cutoffs of market capitalization for small and large stock quintiles are based on the stratification each day using the 3,000 largest U.S. stocks.

[c] For the Nasdaq stocks in the right four columns, cutoffs of B / M for growth and value stock quintiles are based on the stratification each day using the 3,000 largest U.S. stocks.

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Table 5. Panel A, continued

Bottom Left and Top Right Corner Cells in Each 3x3 Matrix

Sample:	3,000 U.S. Stocks, 1996-2008					Nasdaq Stocks, 1997-2001			
Attention Proxy:	VOL_{t-1}					$Retail_NetBuy_{t-1}$			
Return Measure:	cto(%)		otc(%)			cto(%)		otc(%)	
Cell in 3x3 Matrix :	Bottom Left	Top Right	Bottom Left	Top Right		Bottom Left	Top Right	Bottom Left	Top Right
Base Case (Table 3)	.17	.02	-.17	-.03		.34	.02	-.42	-.09
Newey-West t-ratio	11.9 **	3.5 **	-5.6 **	-1.3		11.3 **	1.0	-6.8 **	-1.8
11. Momentum Losers (Low Quintile, 6-mo Return)	.32	.06	-.31	-.08		.48	.09	-.61	-.17
	14.7 **	6.2 **	-8.0 **	-3.0 **		12.5 **	3.2 **	-7.9 **	-2.7 **
12. Momentum Winners (High Quintile, 6-mo Return)	.14	.03	-.13	-.04		.26	.02	-.25	-.12
	9.0 **	3.4 **	-4.0 **	-1.7		8.5 **	1.3	-3.8 **	-1.9
13. Mondays	.20	.04	-.35	-.11		.37	.05	-.66	-.30
	5.8 **	3.0 **	-5.4 **	-2.6 **		5.4 **	1.0	-4.8 **	-2.7 **
14. Other WeekDays	.16	.02	-.12	-.01		.33	.02	-.36	-.04
	10.3 **	2.5 *	-3.8 **	-.2		9.9 **	.7	-5.6 **	-.8
15. 1996 - 2000[d]	.25	.04	-.24	-.03		.37	.05	-.46	-.12
	10.2 **	4.1 **	-5.0 **	-1.4		11.4 **	2.2 *	-6.9 **	-2.2 *
16. 2001 - 2004[e]	.16	.00	-.22	.01		.17	-.11	-.21	.03
	6.4 **	.3	-4.1 **	.5		2.5 *	-2.2 *	-1.4	.3
17. 2005 - 2008	.13	.03	-.16	-.07		–	–	–	–
	5.2 **	2.0 *	-2.7 **	-1.5		–	–	–	–

* indicates statistical significance at the .05 level, and ** indicates significance at the .01 level.

[d] Data on $Retail_NetBuy_{t-1}$ are unavailable before 1997, so the entries in this row for this proxy are for 1997-2000.

[e] Data on $Retail_NetBuy_{t-1}$ are unavailable after 2001, so the entries in this row using this proxy are for 2001 only.

Panel B. Fama-MacBeth Regression Analysis of Model (1)

Sample:	3,000 U.S. Stocks, 1996-2008			Nasdaq Stocks, 1997-2001	
Attention Proxy:	VOL_{t-1}			$Retail_NetBuy_{t-1}$	
Dependent Variable:	cto(%)	otc(%)		cto(%)	otc(%)
Attention	.179	-.272		.218	-.178
	4.8 **	-9.0 **		15.5 **	-6.0 **
Institutional Ownership	-.035	.093		-.113	.335
	-2.5 *	4.4 **		-4.0 **	3.8 **
Firm Size	.024	.159		.010	.309
	0.7	11.9 **		0.3	8.5 **
Avg # Firms / Day	2,342	2,342		1,013	1,005
Avg Adjusted R^2	.045	.045		.021	.015

* indicates statistical significance at the .05 level, and ** indicates significance at the .01 level.

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